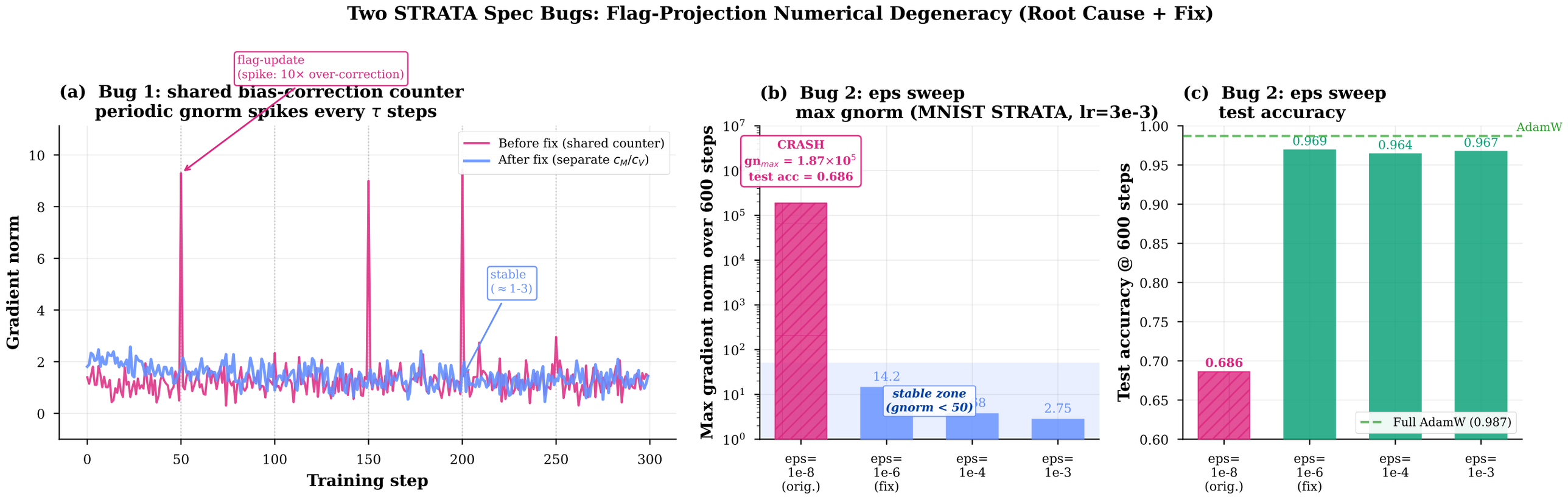

Two STRATA Spec Bugs: Flag-Projection Numerical Degeneracy (Root Cause + Fix)
(a) Bug 1: shared bias-correction counter
periodic gnorm spikes every τ steps
flag-update
(spike: 10× over-correction)
Before fix (shared counter)
After fix (separate c_M/c_V)
stable
(≈ 1-3)
Gradient norm
Training step
(b) Bug 2: eps sweep
max gnorm (MNIST STRATA, lr=3e-3)
Max gradient norm over 600 steps
CRASH
gn_{max} = 1.87×10^5
test acc = 0.686
14.2
stable zone
(gnorm < 50)
2.75
eps=
1e-8
(orig.)
eps=
1e-6
(fix)
eps=
1e-4
eps=
1e-3
(c) Bug 2: eps sweep
test accuracy
Test accuracy @ 600 steps
AdamW
0.686
0.969
0.964
0.967
Full AdamW (0.987)